



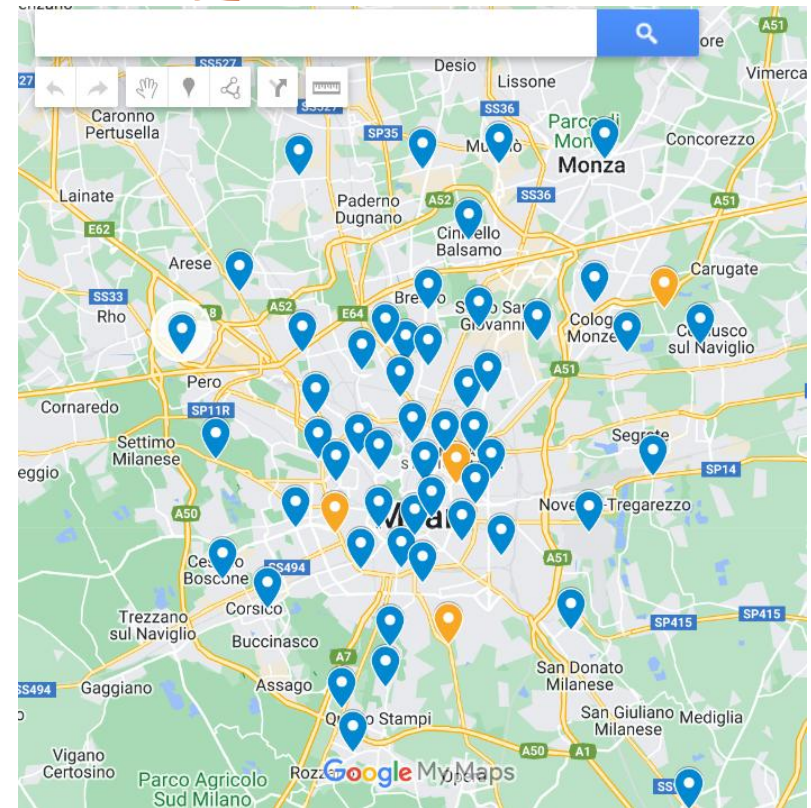
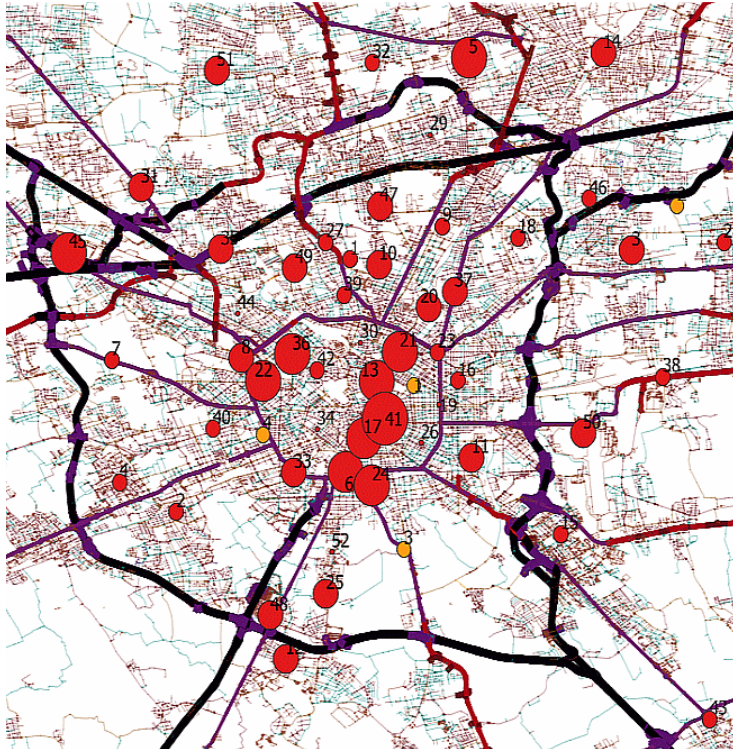
IFORS 2026
The 24th Conference of the International Federation
of Operational Research Societies
12-17 JULY
VIENNA, AUSTRIA

Bootstrap Enhanced Scenario Optimization for Two-Echelon Logistics, a Case Study in Large- Scale Retail

Vittorio Maniezzo, *University of Bologna*

Livio Fenga, *University of Exeter*

The case study



- 52 retailer stores
- 4 DCs: one big, one smaller, 2 satellites close to city center

Allocation for Christmas season! Preliminary analysis during summer.

The first case study

Large-scale organized distribution is often structured according to a two-echelon model, in which distribution centers (DCs) are used to store and forward goods to replenish the inventories of the final retailer's stores.

Stores are typically uniquely allocated to DCs for products of the same line, and predictive knowledge of future demand is essential to optimize the distribution process.

In the real-world case study:

- city retailer store demands were predicted based on individual sales time series and
- used to optimize the allocation of stores to DCs in a tactical planning context.

Our problem required dimensioning multiple DCs and assigning stores to them to meet expected demand during a critical period.

Robust optimization proposes a deterministic reformulation considering worst-case outcomes within a defined uncertainty set.

Distributionally Robust Optimization (DRO) seeks a solution most resilient against the worst-case probability distribution for uncertain parameters, typically within an **ambiguity set** (U): *a set of probability distributions* that the true, unknown distribution of the parameters is assumed to belong to.

DRO **optimizes the worst-case performance** over all distributions within this set. Mainstream DRO approaches define this ambiguity set in different ways.

- *Delage and Ye* (2010) define the set based on **known moment constraints** (e.g., bounds on the mean and covariance) of the uncertain parameters.
- *Ben-Tal et al.* (2013) construct ambiguity sets centered at the empirical distribution using **discrepancy measures based on ϕ -divergences**.
- *Esfahani and Kuhn* (2018) define the ambiguity set as a **Wasserstein ball around the empirical distribution**, yielding tractable convex (often linear) reformulations

Scenario-Based Optimization associates a probability to different parameter realizations (scenarios), and optimizes decisions across all scenarios.

Sample Average Approximation (SAA) is the special case where scenarios are independent and identically distributed realizations, each assigned equal weight $1/N$. Expectation can be replaced by mean, and the stochastic problem is transformed into a deterministic equivalent.

Bertsimas and Kallus (2020) generalize SAA by replacing uniform scenario weights with **covariate-dependent weights** (e.g., via k-nearest), yielding a conditional approximation of the distribution given observed side information.

A **distinct line of work** addresses the same challenge — uncertain parameters estimated from data — from the learning side rather than the distributional side.

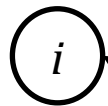
Elmachtoub and Grigas (2022) propose **Smart Predict-then-Optimize** (SPO), training a prediction model with a loss function that measures the resulting decision error, rather than prediction error.

Toward a formulation

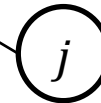


IFORS 2026
The 24th Conference of the International Federation
of Operational Research Societies
12-17 JULY
VIENNA, AUSTRIA

Server i
Capacity Q_i
Unitary storage cost d_i



if j is serviced from i , then $x_{ij} = 1$



Client j
Total request req_j
Allocation cost to i c_{ij}
Amount serviced by i q_{ij}

$$z(D) = \min \quad \sum_{i \in I} \sum_{j \in J} (c_{ij} x_{ij} + d_i q_{ij}) \quad (1)$$

$$\textit{subject to} \quad \sum_{i \in I} q_{ij} = req_j \quad j \in J \quad (2)$$

$$\sum_{j \in J} q_{ij} \leq Q_i \quad i \in I \quad (3)$$

$$\sum_{i \in I} x_{ij} \leq b_j \quad j \in J \quad (4)$$

$$q_{ij} \leq req_j x_{ij} \quad i \in I, j \in J \quad (5)$$

$$x_{ij} \in \{0,1\} \quad i \in I, j \in J \quad (6)$$

$$q_{ij} \in \mathbb{Z}_0^+ \quad i \in I, j \in J \quad (7)$$

$$z(H) = \min \quad \sum_{i \in I} \sum_{j \in J} (c_{ij} x_{ij} + d_i q_{ij}) \quad (1)$$

$$\textit{subject to} \quad \sum_{i \in I} q_{ij} = \rho_j \quad j \in J \quad (2)$$

$$\sum_{j \in J} q_{ij} \leq Q_i \quad i \in I \quad (3)$$

$$\sum_{i \in I} x_{ij} \leq b_j \quad j \in J \quad (4)$$

$$q_{ij} \leq \rho_j x_{ij} \quad i \in I, j \in J \quad (5)$$

$$x_{ij} \in \{0,1\} \quad i \in I, j \in J \quad (6)$$

$$q_{ij} \in \mathbb{Z}_0^+ \quad i \in I, j \in J \quad (7)$$

$$z(\text{SB}) = \min \sum_{s \in S} (p_s \sum_{i \in I} (\sum_{j \in J} (c_{ij} x_{ij} + d_i q_{ij}^s) + M \epsilon_{js})) \quad (1)$$

$$\text{subject to} \quad \sum_{i \in I} q_{ij}^s + \epsilon_{js} = \rho_j^s \quad j \in J, s \in S \quad (2)$$

$$\sum_{j \in J} q_{ij}^s \leq Q_i \quad i \in I, s \in S \quad (3)$$

$$\sum_{i \in I} x_{ij} \leq b_j \quad j \in J \quad (4)$$

$$q_{ij}^s \leq \rho_j^s x_{ij} \quad i \in I, j \in J, s \in S \quad (5)$$

$$x_{ij} \in \{0,1\} \quad i \in I, j \in J \quad (6)$$

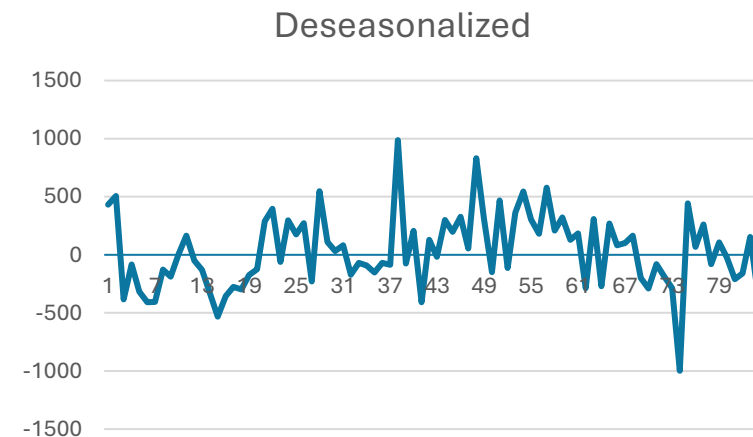
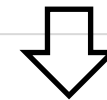
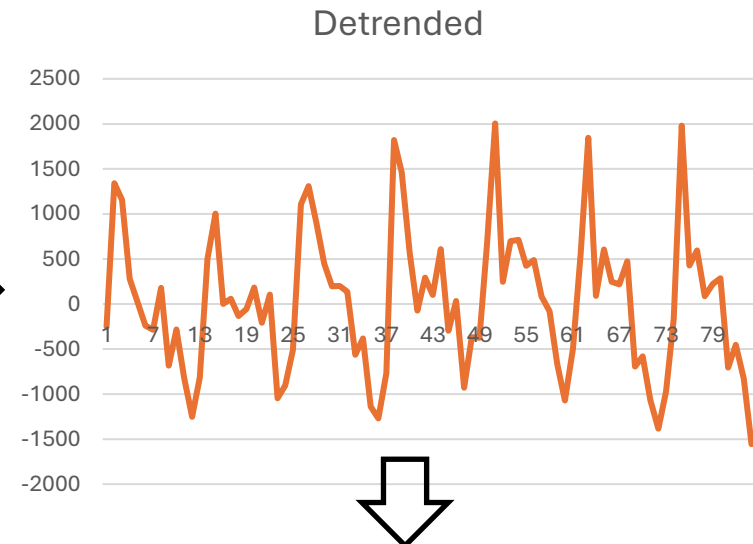
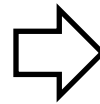
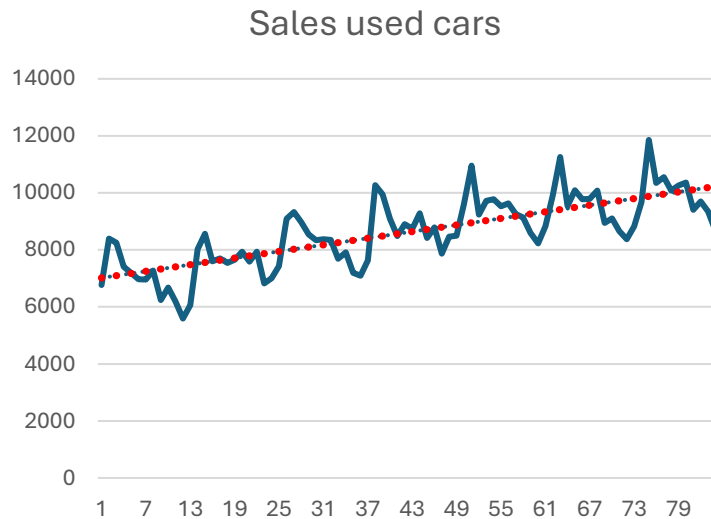
$$q_{ij}^s \in \mathbb{Z}_0^+ \quad i \in I, j \in J, s \in S \quad (7)$$

$$\epsilon_{js} \in \mathbb{Z}_0^+ \quad j \in J, s \in S \quad (8)$$

We
forecast
these

Prewhitening

Economics-related time series can often be effectively decomposed using STL



(almost) white noise

The prewhitened series only retains **residuals unexplained** by the model (i.e. *random shock*).

Residuals are (hopefully) **normally distributed**.

The observed residuals are **realizations of a random variable** whose distribution can be identified.

Using the average and standard deviation of the residual series, I can **generate alternative samples** that are compatible with the overall model, i.e. samples that could have occurred.

Bootstrapping time series

Bootstrapping is non-parametric **resampling technique** used when historical data is available. It makes no assumptions about the underlying probability distribution of the stochastic variables.

The **assumption** of bootstrapping is that the observed historical data sample mimics the characteristics of the true, unknown population distribution.

Bootstrapping **repeatedly draws** samples with replacement from the observed data to generate "*bootstrap samples*". Each sample is **a plausible realization** of what a new data sample from the true population *might* look like.

Autoregressive models are particularly suitable for data series bootstrap set generation.

Bootstrapping

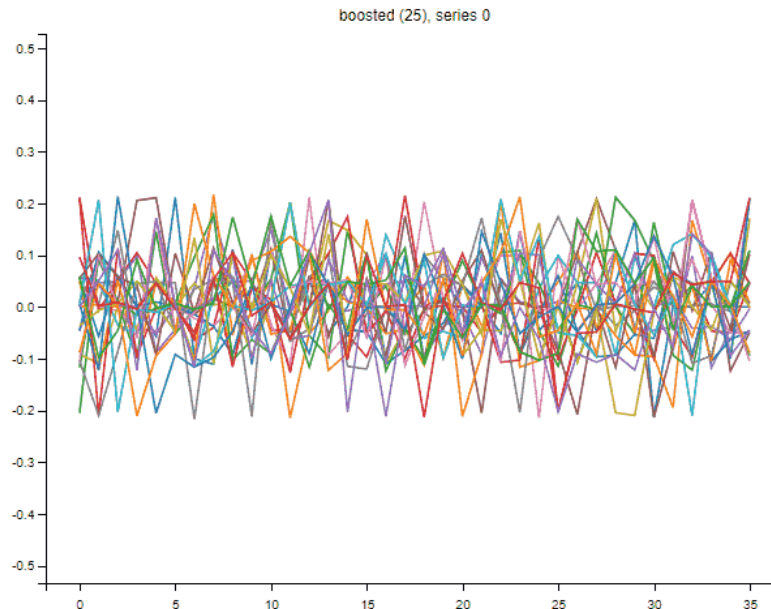
We fit an AR(p) model on each series

We extracted **residuals** ε_t , which were *sufficiently* (?) stationary

We resampled the residuals in ε_t^* and simulated new series as:

$$x_t^* = \phi_1 x_{t-1}^* + \dots + \phi_p x_{t-p}^* + \varepsilon_t^*$$

We bootstrapped the residual series and reconstructed request series.



Maximum entropy bootstrap

We were working on **short data series**, that often produce biased estimates.

The resampling process was guided according to **Maximum Entropy Bootstrap** (MEB, Vinod 1993), associating a probability with each outcome and **maximizing an information entropy** function defined over the resampled data, **subject to constraints** reflecting statistics computed on the original data.

This guarantees that the generated bootstraps are the **least biased**, i.e., the **most non-committal estimates** consistent with the known constraints, i.e., with the observed data.

Objective-Augmented Maximum Entropy Bootstrap (OA-MEB)



IFORS 2026
The 24th Conference of the International Federation
of Operational Research Societies
12-17 JULY
VIENNA, AUSTRIA

Standard MEB defines a probability distribution **maximizing entropy subject to constraints** derived from statistics computed on the empirical dataset.

It can be formulated in a Lagrangian fashion as

$$L(p, \lambda, \mu) = S(p) + \lambda \left(1 - \sum_{i=1}^n p_i \right) + \mu \left(\bar{f} - \sum_{i=1}^n p_i f_i \right)$$

We added to the formulation an **auxiliary objective-related component**, representing scenario-wise quantities that are relevant to downstream cost functions

$$L_{OA}(p, \lambda, \mu, \gamma) = S(p) + \lambda \left(1 - \sum_{i=1}^n p_i \right) + \mu \left(\bar{f} - \sum_{i=1}^n p_i f_i \right) + \gamma \left(\sum_{i=1}^n p_i g_i \right)$$

This **produces synthetic datasets** that reflect the uncertainty of the original data but also adhere to the specific objectives and constraints

BESO: Bootstrapped Enhanced Scenario Optimization



IFORS 2026
The 24th Conference of the International Federation
of Operational Research Societies
12-17 JULY
VIENNA, AUSTRIA

Fit an AR(p) model on each OA-MEB series, independently.

Each model is used to generate a scenario $s \in S$, specifically, a realization ρ_j^s of the forecasted requests of customer $j \in J$ according to scenario S .

All scenarios are deemed equiprobable.

The resulting deterministic equivalent SB formulation is passed on to a MIP solver.

The solution details the server i that each client j will be associated with, as well as the amount q_{ij} it will request from the server.

It is therefore possible to estimate the actual capacity requested from each server.

MEB bootstrap sets were aggregated (bagged) to obtain forecasts.

Besides MEB bagging we also considered alternative forecasts modules based on:

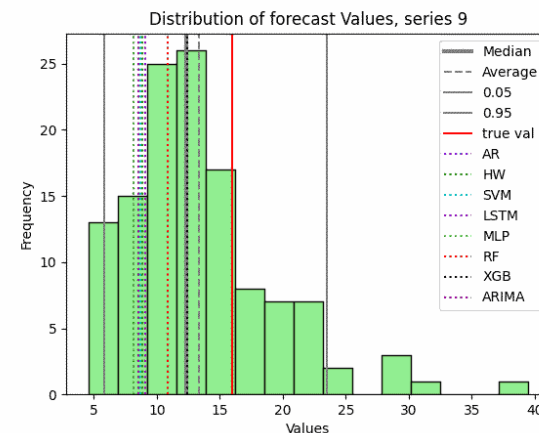
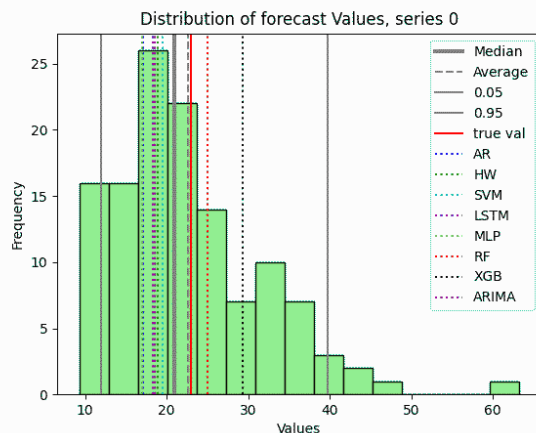
- AR
- Holt-Winters
- SARIMA
- Multilayer perceptrons
- LSTM networks
- Random forests
- XGboost

Results, forecasts

MEB based forecasting. Average rankings for different boostset sizes.

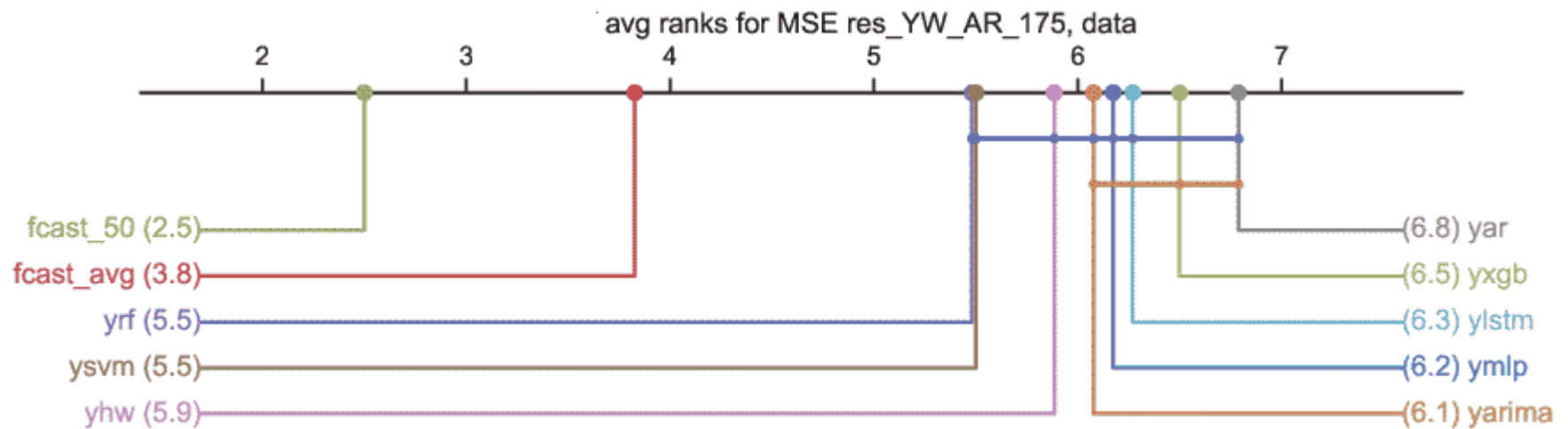
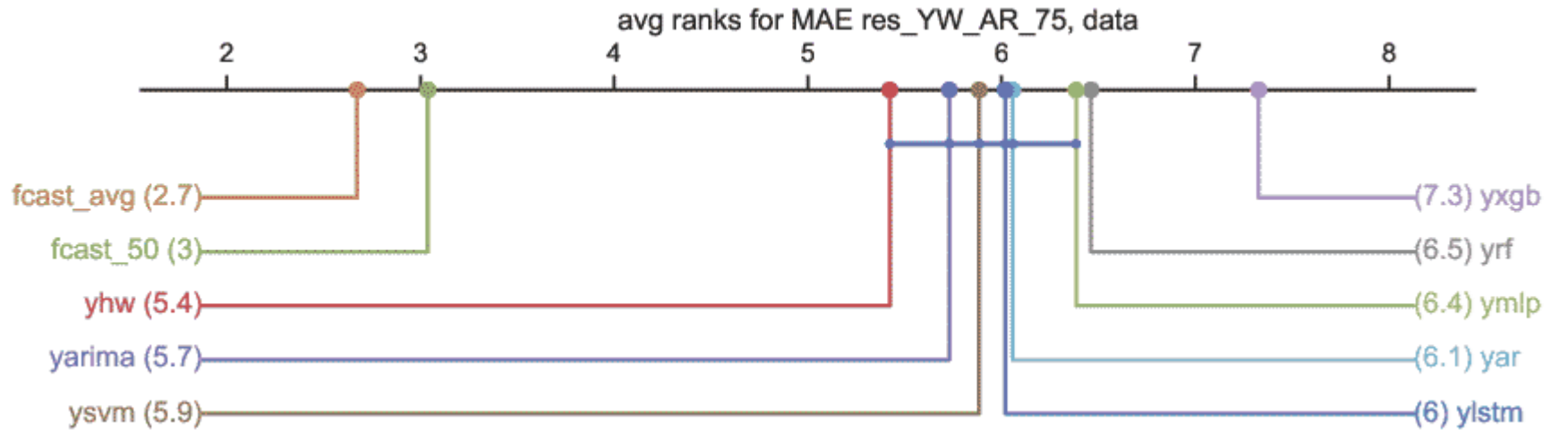
	YW_AR_75			YW_AR_125			YW_AR_175		
	MAE	MSE	bias	MAE	MSE	bias	MAE	MSE	bias
fcast_avg	2.67	2.67	0.08	2.58	2.58	0.05	3.83	3.83	0.04
fcast_50	3.04	3.04	0.06	3.19	3.19	0.04	2.50	2.50	0.03
yhw	5.42	5.42	0.08	6.06	6.06	0.04	5.88	5.88	0.03
varima	5.73	5.73	0.05	6.04	6.04	0.03	6.08	6.08	0.02
ysvm	5.88	5.88	0.08	5.50	5.50	0.05	5.50	5.50	0.03
ylstm	6.02	6.02	0.06	5.52	5.52	0.04	6.27	6.27	0.02
var	6.06	6.06	0.04	6.54	6.54	0.02	6.79	6.79	0.02
ymlp	6.38	6.38	0.05	5.98	5.98	0.03	6.17	6.17	0.02
yrf	6.46	6.46	0.11	6.31	6.31	0.06	5.48	5.48	0.04
yxgb	7.33	7.33	0.12	7.29	7.29	0.07	6.50	6.50	0.05

Table 1 Average rankings for different boostset size.



Results, forecasts

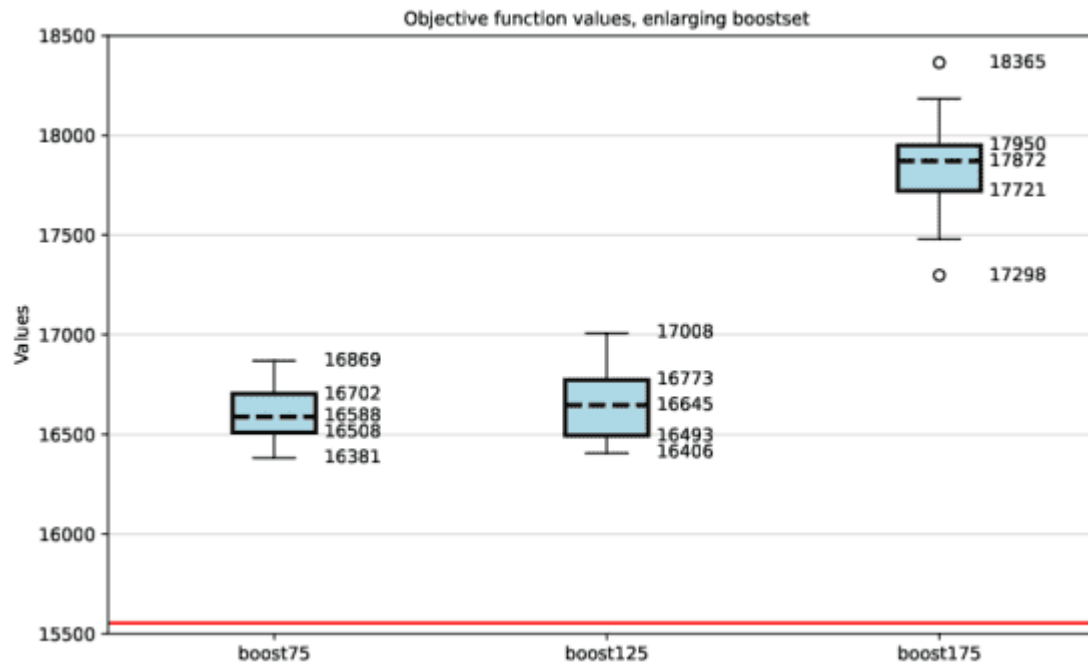
Critical difference diagrams (Fawaz et al., 2019)



Results, optimization



IFORS 2026
The 24th Conference of the International Federation
of Operational Research Societies
12-17 JULY
VIENNA, AUSTRIA



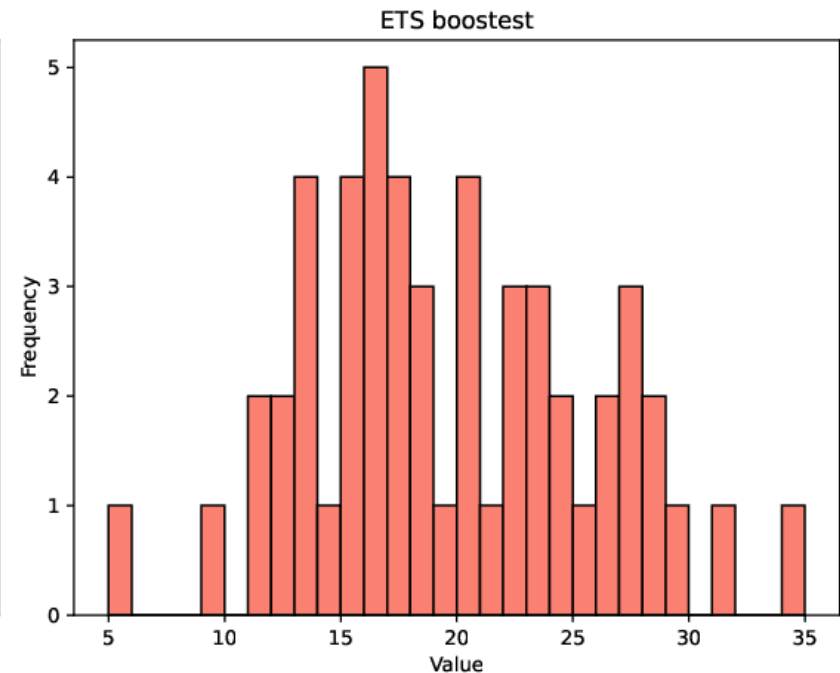
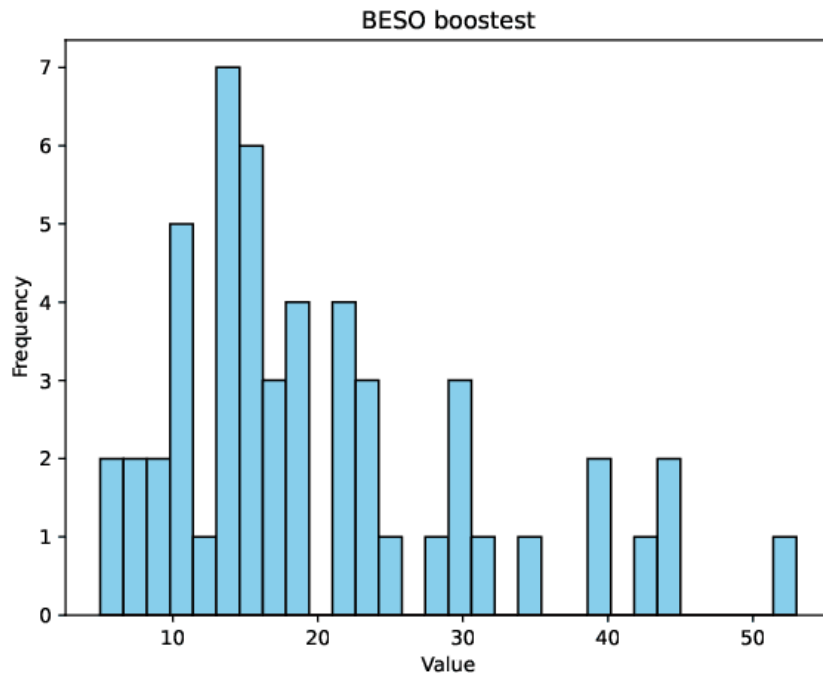
The cost is essentially stable in the cases of the 75 and 125 series sets but increases significantly in the case of the 175 boostset.

This is due to some extreme scenarios appearing in the forecasts.

All instances were feasible, but 7 out of 35 tests still had a gap between lower and upper bound after 600 seconds of CPU time.

Results, comparison

BESO vs. distribution-based, gaussian derived from confidence intervals of Holt-Winters ETS forecasts.



Results, comparison

Comparing against a distribution-based optimization (Holt-Winters, confidence intervals)

	<i>No rental cost</i>		<i>With rental cost</i>	
	BESO	ETS	BESO	ETS
Assign. Cost	17781	15423	5446720	4378380
Optimality gap %	0	0	16.07	0.69
Infeasibilities	0	1	0	1
t.Cpu	33	29	7200	7200
Allocated capacities:				
Server 0	67 / 120	119 / 120	82 / 120	116 / 120
Server 1	571 / 1000	430 / 1000	532 / 1000	428 / 1000
Server 2	257 / 300	301 / 300	257 / 300	303 / 300
Server 3	128 / 180	173 / 180	152 / 180	176 / 180

BESO is a framework for **deriving predictive prescriptions from relevant data series**. It integrates statistically grounded forecasting with scenario-based optimization by generating future scenarios via bootstrapped autoregressive time series, while **incorporating feedback from the optimization stage**.

Bootstrapped forecasting identifies a probability distribution of future retailer requests, supporting **data-driven scenario generation** while considering both costs and forecast uncertainty.

The validity of the BESO approach **extends beyond the case study**.

Integrating forecasting and optimization within a unified empirical loop enhances **stability**, operational **effectiveness** and result **justification**.